

Your grade: 100%

Learning Check - Random Sampling

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Review Learning Objectives

Instructions

1. Which of the following best describes random sampling?

1. Which of the following best describes random sampling?

1 / 1 point

- ☐ Selecting individuals from a population based on specific criteria.
 ☒ Choosing individuals from a population without any pattern or bias.
 ☐ Sampling only a small portion of the population to save time and resources.
 ☐ Selecting individuals from a population using convenience sampling methods.
- Sep 30, 11:59 PM IST 4 left (5 attempts every 24 hours)

Correct

Random sampling involves selecting individuals from a population in such a way that every member of the population has an equal chance of being chosen. This ensures that the sample is representative of the population and helps to minimize bias in the sample.

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Your grade

To pass you need at least 80%. We keep your highest score.

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- ☐ Probability distribution
- ☐ Confidence interval
- ☒ Estimation error
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☒ Correct

This is the correct option. Estimation error refers to the difference between the estimated value (such as $\bar{x}$) and the true value of the population parameter (μ) that the estimate is attempting to capture.

3. Estimating the average height of women in Philadelphia seemed like a lot of work, so we instead decided to estimate the average height of female Penn students.

1 / 1 point

We measured 5 stu
mean of our data?

65.4

✓ **Correct**
The sample mean $\bar{X} = (X_1 + X_2 + \dots + X_n)/n$ so for our 5 samples, $\bar{X} = (64 + 66 + 67 + 63 + 67)/5 = 65.4$.